FEBIN C S

Embedded Systems Engineer | Firmware Development & PCB Design

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PROFESSIONAL SUMMARY

Embedded systems engineer who builds working hardware end to end: circuit and PCB design in EasyEDA and KiCad, firmware in Embedded C on ESP32, Arduino and STM32, and the bring-up and debugging in between. Led the teams behind a laser link that carries text and audio and an ESP32 hydroponics controller; IoT competition winner and IEDC Summit 2026 trainer.

TECHNICAL SKILLS

Microcontrollers & Boards: ESP32, ESP8266 / NodeMCU, Arduino (UNO, Nano), STM32, Raspberry Pi, Raspberry Pi Pico
Interfaces & Protocols: UART, I2C, SPI, single-wire, Wi-Fi (TCP sockets), Bluetooth, ESP-NOW, LoRa (SX1278), MQTT
PCB Design: EasyEDA (primary), KiCad, Proteus; boards with and without microcontrollers, schematic to Gerbers, etching
Hardware & Debugging: Soldering, rework, board bring-up; multimeter and oscilloscope fault finding; appliance board repair
Programming & Tools: Embedded C, C, Python, STM32CubeIDE, Arduino IDE, MATLAB, Blynk, Linux

EMBEDDED PROJECTS

LasE Wave - Laser Communication System | Arduino Nano, 650 nm laser, IRF520N, LA6283; B.Tech final project, 2025

- Led a team of four and owned the circuit, hardware and firmware of a free-space optical link carrying keypad-typed text and audio over 1 m: an Arduino Nano switches a 650 nm laser through an IRF520N MOSFET driver.
- Wrote the firmware: multi-tap text entry on a 4x4 keypad, Morse-coded on-off keying of the laser, and a receiver that times pulses from a solar-panel photodetector to decode characters onto an I2C LCD.
- Designed and had fabricated the carrier PCB (Nano socket, voltage regulators, text/audio mode switch, power input, headers) and built the audio path around a 45 dB LA6283 amplifier.

HydroGreens - IoT Hydroponics Controller | ESP32, DHT11, TDS, MQ9, water-level sensor, Blynk; team lead, 2024

- Led a team of four and built an ESP32 controller that reads temperature, humidity, TDS, gas and water level, refills the reservoir when the level drops and doses nutrients every 24 hours, with a grow light, exhaust fan and air pump.
- Streamed live readings to a Blynk dashboard; in a 7-day trial the microgreens grew faster than a batch in open conditions.

Device-to-Device Communication Kit | Arduino UNO, ESP32, NodeMCU, SX1278 LoRa; workshop material, 2026

- Built paired sender and receiver firmware for eight links: UART, I2C, SPI, single-wire, Wi-Fi, Bluetooth, ESP-NOW and 433 MHz LoRa. Includes a register-level SPI slave, a bit-banged single-wire link with microsecond timing, and callback-driven I2C and ESP-NOW receivers.

Other builds | Arduino, STM32, Raspberry Pi

- Laser Game (Lazer Maze Runner): Arduino laser-and-sensor game with real-time scoring; 100+ players at two fests.
- STM32 (STM32CubeIDE): fundamentals-level builds with sensor interfacing and board-to-board communication.
- Seido Tracker (diploma project, team of 6): Raspberry Pi reading runners' RFID tags to count laps and record finish times.

KEY ACHIEVEMENTS

- Workshop trainer at IEDC Summit 2026 (Kerala Startup Mission; billed as Asia's largest for aspiring entrepreneurs, 10,000+ attendees): ran hands-on sessions on Device-to-Device Communication and on PCB Design, where each participant drew a two-transistor LED flasher in EasyEDA and etched it as a single-sided board.
- First Prize, Internet of Things competition, SAE India Collegiate Club, GEC Palakkad (TIER 1 events, Aug 2024).
- Hired by Jyothi Engineering College to redeploy the laser game from Invento 2023 at its Tharang 2.0 fest (Nov 2023).
- Appreciated for guiding STM32-based project development at the Firmware Frenzy workshop (Mar 2025).

EXPERIENCE

Academic Executive, Keltron Knowledge Centre, Thrissur — Nov 2025 - Present

- Run hands-on electronics and embedded labs; help students build and debug their hardware and firmware.

Junior Robotics Instructor, Tinkertodz

- Taught robotics and electronics through builds: Arduino programming, sensor integration and mechanical design.

LEADERSHIP

- Co-founder and Project Head (2024-25), Tecknow Club, the Dept. of ECE's student technical community.
- Quality and Operation Lead, IEDC GEC Palakkad, 2024-25; Coordinator, Technical Events, Invento 2023.

EDUCATION

B.Tech, Electronics & Communication Engg., GEC Palakkad (course completed; degree awaited) — 2022 - 2025
Diploma, Electronics & Communication Engg., Govt. Polytechnic College, Kunnamkulam (First Class) — 2018 - 2021
Languages: English, Malayalam